Data Sheet

EXPERIMENT NO. 1(a)

Title: - Study of Ripple Factor and Voltage Regulation Characteristics of Half-Wave Rectifier Without Capacitor Filter (using Tinkercad software circuit).

Apparatus Required: -

Sl. No.	Apparatus / Component	Specification	Quantity
1	Function Generator	10.0Hz, 5V, Sine wave	1
2	Oscilloscope	10.0V Scale	1
3	Diode	Standard Silicon	1
4	Resistor	1K, 10K, 100K (Ω)	3
5	Bread board	Small	1
6			

Important Formulae: -

Ripple Factor: -

Ripple factor is defined as the ratio of RMS value of AC component to the DC component present in the rectifier output.

For Half-Wave Rectifier (Without Filter)

r = 1.21

Experimental Ripple Factor: -

r = Vac / Vdc

Where:

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Experiment No: 1(a)

⊙ Title :- Study of Ripple factor and voltage Regulation and characteristics of Half Wave Rectifier Without Capacitor.

⊙ Theory :- A half wave rectifier is an electronic circuit that converts an AC voltage into a pulsating DC voltage by allowing only one half-cycle of the AC input to pass through the diode. The diode conducts during the positive half-cycle producing a pulsating DC output. The Output of the rectifire contains both AC Component and AC Component.

Ripple factor is defined as the ratio of the RMS value of the AC Component to the DC Component present in the rectifire output.

The theoritical ripple factor is $r = 1.21$

The experimental ripple factor is : $r = V_{ac}/V_{dc}$

V_{ac} = AC ripple voltage

V_{dc} = DC ripple voltage.

Rectifience efficiency is the ratio of DC power delivered to the load to AC power Supplied to rectifire, expressed as percentage

$$\eta = (P_{dc}/P_{ac}) \times 100\%.$$

For half rectifiere η = 40.6%.

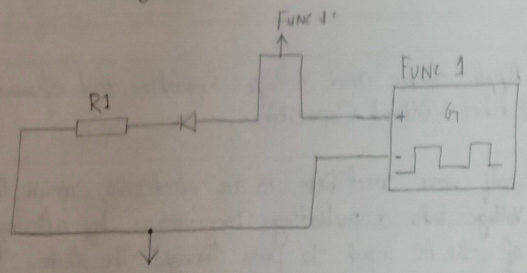
Voltage Regulation indicates the change in output voltage when the load changes from no load to full load.

$$\text{Voltage Regulation (\%)} = [(V_{NL} - V_{FL})/V_{FL}] \times 100.$$

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⊙ Circuit Diagram:-



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Where VNL = No load voltage
VFL = Full-load voltage.

⊙ Component Required:-

Apparatus	Specification	Quantity
1) Function generator	10Hz, DC offset = 0V, Amplitude = 5V, Sine	1
2) Oscilloscope	1.00 per div, V.S = 10V fullw P	1
3) Breadboard	Half Sized with 3000Ω units	1
4) Resistor	1KΩ, 10KΩ, 100KΩ	1
5) Diode	1N4007 type	1
6) Connecting Wires	Solid Core hookup wires for breadboard Connection.	6

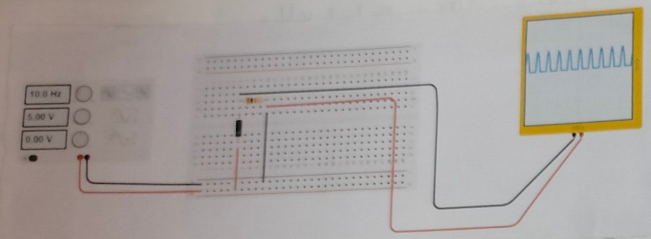
⊙ Observation Table:-

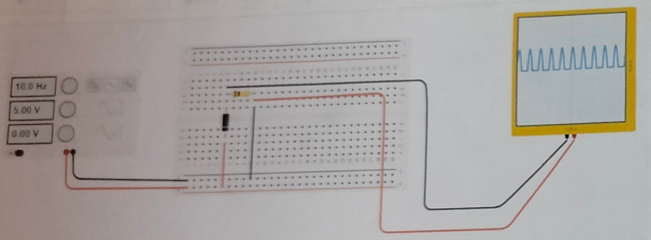
1) Table 1: Without Capacitor filter =)

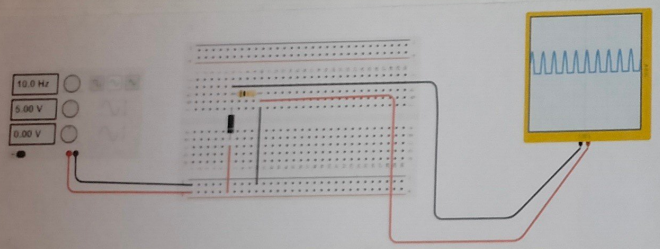
Sl.No	Resistance (KΩ)	Vdc (V)	Vac (V)	Ripple Factor
1.	100 KΩ	0.36	0.771	1.21
2.	10KΩ	0.054	1.152	1.21
3.	1KΩ	1.378	2.75	1.21

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⊙ Calculation for Table 1

(i) When Amplitude = 5V:

$V_{dc} = V_m/\pi = 2/\pi = 0.633\,V$

$V_{rms} = V_m/2 = 1\,V$

$V_{ac} = \sqrt{V_{rms}^2 - V_{dc}^2} = 0.771\,V$

Ripple factor = $0.771/0.636 \approx 1.21$

(ii) When Amplitude = 7V:

$V_{dc} = V_m/\pi = 3/\pi = 0.954\,V$

$V_{rms} = V_m/2 = 1.5V$

$V_{ac} = \sqrt{V_{rms}^2 - V_{dc}^2} = 1.157\,V$

R.F = $1.157/0.954 \approx 1.212$

(iii) When Aptitude = 10V:

$V_{dc} = V_m/\pi = 1.31/\pi = 1.362$

$V_{rms} = V_m/2 = 2.15\,V$

$V_{ac} = \sqrt{V_{rms}^2 - V_{dc}^2} = 2.751$

R.F = $2.751/1.362 = 1.212$

(iv) Table 2: Voltage Regulation =>

Condition	Output Voltage (V)
No load	0.771 V
Full load	0.63V

Voltage Regulation = $[(V_{NL} - V_{FL})/V_{FL}] \times 100$

$= [(0.771 - 0.63)/0.63] \times 100$

$= 21.6\%$

⊙ Conclusion: The half-wave rectifire without a capacitor filter was succesfully studied using the given factor was found to be approximately 1.21-123, which is close to the theoritical value of 1.21. The voltage regulation was calculated to be approximately 21.6%. Confirming the change in output voltage form no-load to full-load condition.

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Data Sheet

EXPERIMENT NO. 1(b)

Title: - Study of Ripple Factor and Voltage Regulation Characteristics of Half-Wave Rectifier With Capacitor Filter (using Tinkercad software circuit).

Apparatus Required: -

Sl. No.	Apparatus / Component	Specification	Quantity
1.	Oscilloscope	Standard	2
2.	Function Generator	10 Hz, 5V amplitude	1
3.	Resister	100Ω, 1 KΩ	1
4.	Diode	Silicon Diode	1
5.	Capacitor	1μf, 1000 μf	1
6.	Breadboard	Small	1
7.	Jumper Wire		Required

Important Formulae: -

Peak to peak ripple voltage $Vr = \frac{Vm}{fRC}$

$V_{rms} = \frac{V_m}{2\sqrt{3}}$

Average DC output voltage with capacitive filter $Vdc = Vm - \frac{Vm}{2fRC}$

Ripple factor with capacitive filter $\gamma = \frac{1}{2\sqrt{3}fRC}$

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Experiment NO- 1(b)

⊙ Title:- Study of Ripple Factor and Voltage Regulation Characteristics of Half-Wave Rectifier With Capacitor filter.

⊙ Theory:- A full wave rectifier is an electronic circuit used to convert an AC input voltage into a positive pulsating DC output voltage. In a full wave rectifier both the positive and negative half-cycles of the AC input are utilized to produce the output.

For a full wave rectifier without a capacitor filter, the output contins a DC component along with an AC component, which is known as the ripple component. The RMS value of the output voltage is related to the peak voltage V_m by

$$V_{rms} = V_m/2\sqrt{3}$$

The DC output voltage of a full-wave rectifier is:

$$V_{dc} = V_m - \frac{V_m}{2fRC}$$

The AC Ripple voltage is: $V_{ac} = \sqrt{(V_{rms})^2 - (V_{dc})^2}$

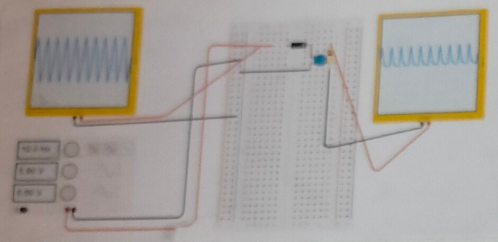
Ripple factor indicates the amount of AC component present in the rectifier output compared with is dc component.

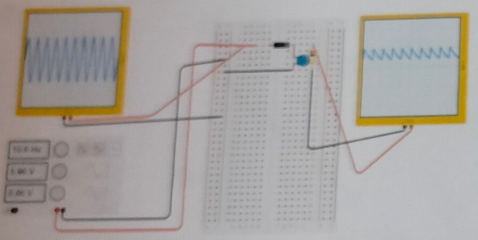
RF (Theoritical) = $\gamma = 1/(2\sqrt{3}fRC)$

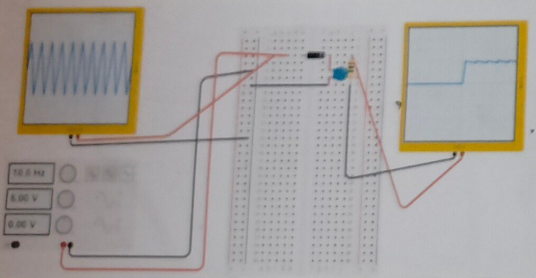
RF (Practical) = $\gamma = V_{ac}/V_{dc}$

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Voltage Regulation represents the change in output voltage when the rectifire is changed from no-load Condition to full load Condition. The no-load output voltage is represented by V_{NL} While the full load Output is represented by V_{FL}.

The voltage regulation = $[(V_{NL} - V_{FL}) / V_{FL}] \times 100\ \%$

where V_{NL} = NO load voltage

V_{FL} = Full load voltage

⊙ Components Required :—

Apparatus	Specification	Quantity
1) Function Generator	10Hz, Offset = 0V, Amplitude 5VA, Sine Wave	1
2) OScillo Scope	2.00 per div, Vs = 10 f up	2
3) Bread Board	Half Sized with 3000 L	1
4) Resistor	1 KΩ, 5 KΩ	1
5) Multimeter	IN4007 type	1
6) Diode	Voltage Regulation	1
7) Capacitor	10 uf, 100 uf	1
8) Connecting Wires	Solid core hookup Wires	10

⊙ Observation Tables :—

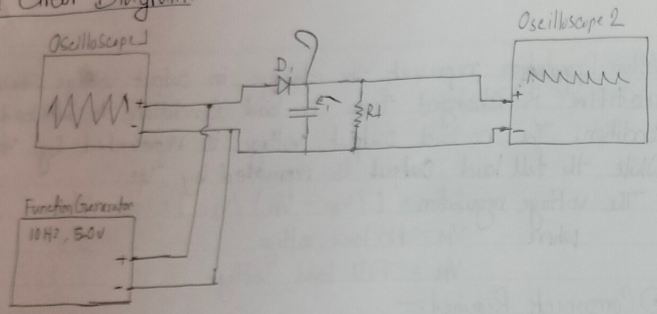
1) Table 1 : With Capacitor Filter ⇒

C (uf)	R_L (KΩ)	V_{dc} (V)	V_{rms} (V)	Ripple factor (practical)	Ripple factor (theoretical)
4) 100	1	2.5V	1.44V	0.576	0.288
2) 1000	1	4.75V	2.144V	0.036	0.018
3) 100	0.1	1.59V	1.99V	1.21	2.88

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Circuit Diagram:-



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Calculation for Table 1:-

① Case 1:- (C = 100uf, R = 1KΩ)

$V_r = \frac{V_m}{fRC} = 0.5$

$V_{rms} = V_r/2\sqrt{3} = 1.44\,V$

$V_{dc} = V_m - \left(\frac{V_m}{2fRC}\right) = 2.5$

R.F (Theory) = $V_{rms}/V_{dc} = 0.288$

R.F (Prac) = $1/2\sqrt{3}fCR = 0.576$

② Case 2 (C = 1000uf, R = 1KΩ)

$V_r = V_r/fRC = 0.92$

$V_{rms} = V_r/2\sqrt{3} = 0.144$

$V_{dc} = V_m - \left(\frac{V_m}{2fRC}\right) = 1.75$

RF (Theory) = $V_{rms}/V_{dc} = 0.128$

RF (Prac) = $1/2\sqrt{3}fCR = 0.036$

③ Case 3: (C = 100uf, R = 0.1KΩ)

$V_r = V_m/fRC = 6V.$

$V_{rms} = V_r/2\sqrt{3} = 1.93$

$V_{dc} = V_m - (V_m/2fRC) = 1.59$

R.F (Theory) = $(V_{rms}/V_{dc}) = 2.88$

R.F (Prac) = $1/2\sqrt{3}fCR = 1.21$

④ Table 2 : Voltage Regulation :-

Condition	Output Voltage (V)
NO-load (V_{NL})	5.00
Full-load (V_{FL})	4.75

Voltage Regulation = $[(V_{NL} - V_{FL})/V_{FL}] \times 100 = [(5 - 4.75)/4.75 \times 100] = 5.26\%$

⑤ Conclusion:- The half-wave rectifire with capacitor filter was Sucessfully studied. The voltage regulation was found to be 5.26% and the practical ripple factor values were found to be close to theoritical values. The experiment shows that ripple factor decreases with increase in Capacitance.

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